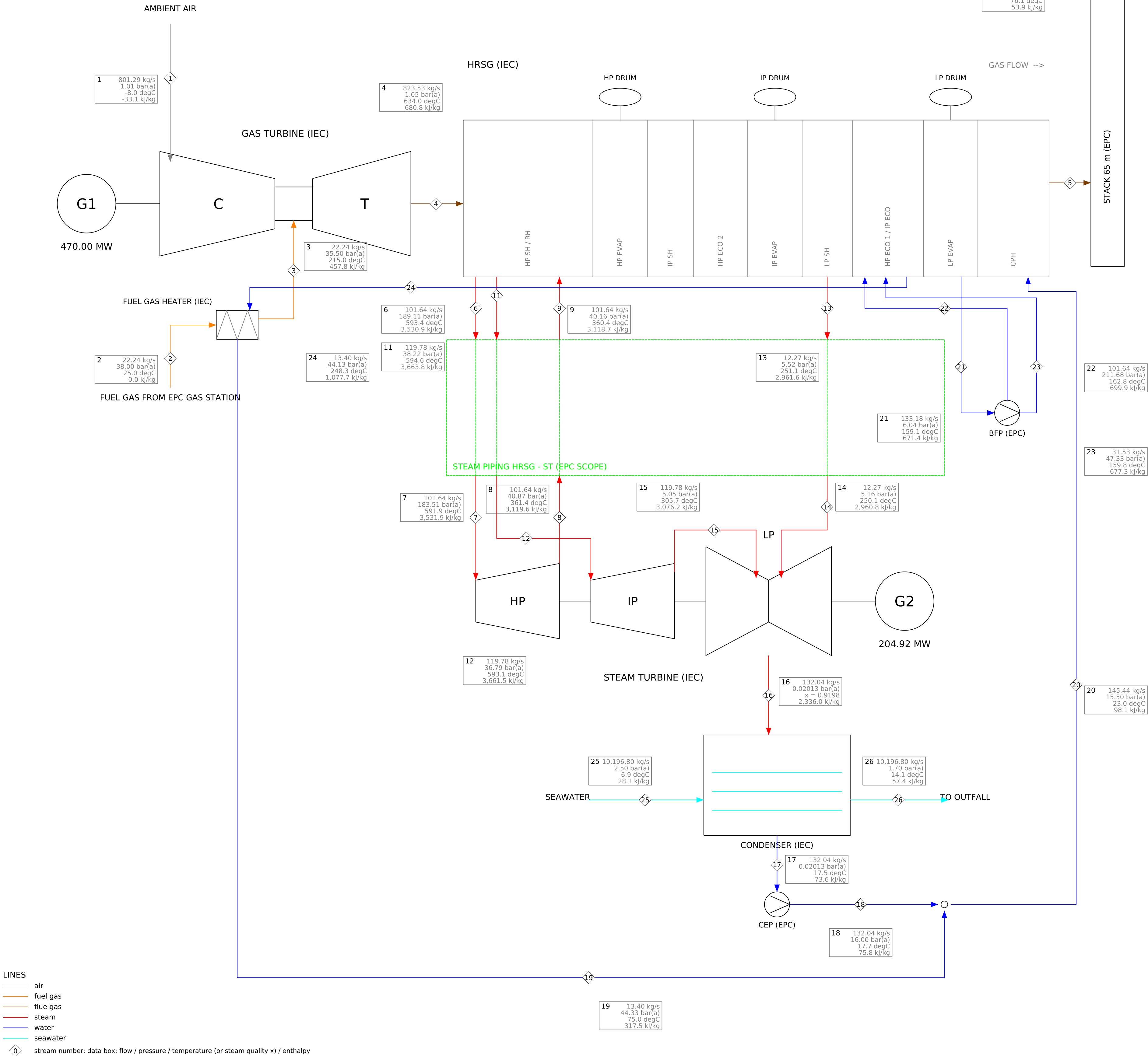




STREAM TABLE - MIN-NG-100

| No | Stream                                                              | kg/s      | bar(a) | degC / x | kJ/kg   | MW*   |
|----|---------------------------------------------------------------------|-----------|--------|----------|---------|-------|
| 1  | Ambient air to GT compressor inlet                                  | 801.29    | 1.01   | -8.0     | -33.1   | -26.5 |
| 2  | Fuel gas at performance gas heater inlet (IEC terminal point IF-01) | 22.24     | 38.00  | 25.0     | 0.0     | 0.0   |
| 3  | Heated fuel gas to GT combustion system                             | 22.24     | 35.50  | 215.0    | 457.8   | 10.2  |
| 4  | GT exhaust gas to HRSG inlet                                        | 823.53    | 1.05   | 634.0    | 680.8   | 560.7 |
| 5  | Flue gas at HRSG outlet to stack                                    | 823.53    | 1.01   | 76.1     | 53.9    | 44.4  |
| 6  | HP steam at HRSG outlet (HP SH outlet)                              | 101.64    | 189.11 | 593.4    | 3,530.9 | 358.9 |
| 7  | HP steam at ST main stop valve inlet                                | 101.64    | 183.51 | 591.9    | 3,531.9 | 359.0 |
| 8  | Cold reheat steam at HP turbine exhaust                             | 101.64    | 40.87  | 361.4    | 3,119.6 | 317.1 |
| 9  | Cold reheat steam at HRSG reheater inlet                            | 101.64    | 40.16  | 360.4    | 3,118.7 | 317.0 |
| 10 | IP superheated steam (joins cold reheat inside the HRSG)            | 18.13     | 41.02  | 329.4    | 3,038.2 | 55.1  |
| 11 | Hot reheat steam at HRSG outlet                                     | 119.78    | 38.22  | 594.6    | 3,663.8 | 438.8 |
| 12 | Hot reheat steam at ST IP stop/control valve inlet                  | 119.78    | 36.79  | 593.1    | 3,661.5 | 438.6 |
| 13 | LP steam at HRSG outlet                                             | 12.27     | 5.52   | 251.1    | 2,961.6 | 36.3  |
| 14 | LP steam at ST LP admission                                         | 12.27     | 5.16   | 250.1    | 2,960.8 | 36.3  |
| 15 | IP turbine exhaust (crossover, internal)                            | 119.78    | 5.05   | 305.7    | 3,076.2 | 368.5 |
| 16 | LP turbine exhaust to condenser (UEEP)                              | 132.04    | 0.0201 | x 0.920  | 2,336.0 | 308.5 |
| 17 | Condensate at hotwell outlet / CEP suction                          | 132.04    | 0.0201 | 17.5     | 73.6    | 9.7   |
| 18 | Condensate at CEP discharge                                         | 132.04    | 16.00  | 17.7     | 75.8    | 10.0  |
| 19 | Fuel gas heater water return to condensate                          | 13.40     | 44.33  | 75.0     | 317.5   | 4.3   |
| 20 | Condensate to HRSG condensate preheater inlet                       | 145.44    | 15.50  | 23.0     | 98.1    | 14.3  |
| 21 | Feedwater from LP drum to boiler feed pumps                         | 133.18    | 6.04   | 159.1    | 671.4   | 89.4  |
| 22 | HP feedwater at BFP discharge to HP economiser                      | 101.64    | 211.68 | 162.8    | 699.9   | 71.1  |
| 23 | IP feedwater at BFP interstage to IP economiser                     | 31.53     | 47.33  | 159.8    | 677.3   | 21.4  |
| 24 | IP economiser water to performance fuel gas heater                  | 13.40     | 44.13  | 248.3    | 1,077.7 | 14.4  |
| 25 | Circulating seawater to condenser inlet                             | 10,196.80 | 2.50   | 6.9      | 28.1    | 286.3 |
| 26 | Circulating seawater from condenser outlet                          | 10,196.80 | 1.70   | 14.1     | 57.4    | 585.1 |

\* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

|                                       |         |        |
|---------------------------------------|---------|--------|
| GT output (generator terminals)       | 470.00  | MW     |
| ST output (generator terminals)       | 204.92  | MW     |
| Gross output                          | 674.92  | MW     |
| Auxiliary loads                       | -9.97   | MW     |
| Transformer losses (GSU, UAT, MV/LV)  | -2.15   | MW     |
| NET OUTPUT (HV terminals)             | 662.80  | MW     |
| Heat input (LHV)                      | 1,056   | MW     |
| Gross heat rate (LHV)                 | 5,633   | kJ/kWh |
| NET HEAT RATE (LHV)                   | 5,736   | kJ/kWh |
| Net efficiency (LHV)                  | 62.76   | %      |
| Fuel LHV (ISO 6976, from composition) | 47.48   | MJ/kg  |
| Fuel gas flow                         | 109,310 | Sm3/h  |
| Circulating water flow                | 36,047  | m3/h   |
| GT load                               | 100.00  | %      |
| PLANT LOAD (NET, % OF SRC-NG-100)     | 107.96  | %      |

CHECKS (guarantees, limits, balances)

OK energy balance CMIX: residual -0.001 MW (tolerance 0.25 MW)  
OK energy balance COND: residual -0.032 MW (tolerance 0.59 MW)  
OK energy balance FGH: residual +0.003 MW (tolerance 0.25 MW)  
OK energy balance GT: residual +0.007 MW (tolerance 1.04 MW)  
OK energy balance HRSG: residual +2.088 MW vs stated casing loss 2.065 MW (tolerance 0.98 MW)  
OK energy balance PIPE: residual +0.274 MW (tolerance 1.15 MW)  
OK energy balance ST: residual -0.003 MW (tolerance 0.83 MW)  
OK gas flow 109,310 Sm3/h = 91.1 % of DP-OWNR-0033 (120,000 Sm3/h)  
INFO net export 662.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case); capability above the capacity, export limited: 640 MW net at the 380 kV revenue meters (TQ-027, TP-E1)

|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| PROJECT  | Alpilla 600 MW Combined Cycle Power Plant                             |
| TITLE    | HEAT AND MASS BALANCE - MIN-NG-100                                    |
| CASE     | Minimum ambient for full capability (-8 degC), natural gas, base load |
| DOCUMENT | ALP-HMB-MIN-NG-100                                                    |
| STATUS   | preliminary - generated by engine hmb v1.1.1                          |
| DATA     | a6baea26d6584942   2026-09-29                                         |
| NOTE     | all values calculated from the database; do not edit                  |